THE ENVIRONMENT AS A STAKEHOLDER

We consider the environment as a stakeholder and an integral part our business and corporate citizenship.

We are firmly committed to operating in a responsible and sustainable manner. Our commitment requires considerations of impacts on the environment at each stage of development.

All activities are conducted safely and reliably to protect the public, minimize environmental impact, safeguard the health and well-being of employees and contractors, and protect our facilities and equipment from damage or loss.

Emissions

Surface Footprint

Risk Mitigation

Asset Retirement

We are firmly committed to maximizing positive impacts on current and future generations and minimizing the impact our operations have on the environment

EMISSIONS

Carbon Sequestration In Midale

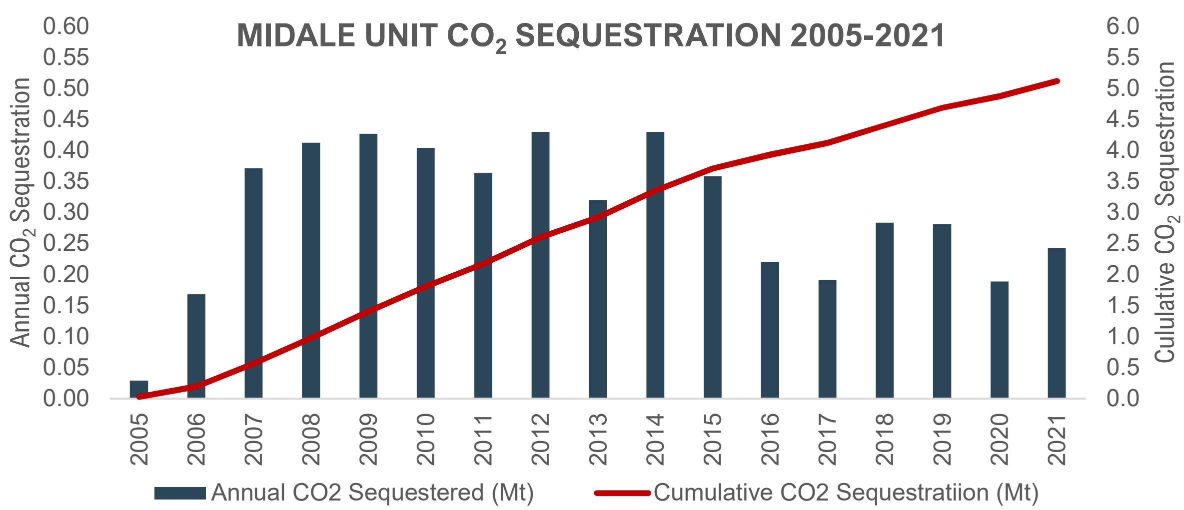
In 2021, Cardinal continued its operation of a best in class, carbon capture and storage project within the Midale unit in southeastern Saskatchewan. Cardinal operates and owns the majority share of the field, where we inject CO_2 from a North Dakota coal gasification project and safely store the CO_2 deep below surface, under layers of impermeable cap rock. This injected carbon dioxide, which would otherwise be in the atmosphere contributing to climate change, serves to increase production from a stable, low decline oil reservoir.

Approximately 242,000 tonnes of CO_2 was injected in the Midale Unit in 2021, a tonnage greater than the CO_2e emissions Cardinal directly emitted from all of our Operations. Cardinal continues to prioritize CO_2 injection as a direct strategy to mitigate climate related risks. Cardinal increased CO_2 injection in 2021, and continues to pursue opportunities to take advantage of our position as being one of the very few organizations that have an operating project that is approved to directly inject CO_2 subsurface.

In 2019, 2020 and 2021, Cardinal injected 1.23 tonnes of CO_2 for every tonne of CO_2e that was directly emitted from our operations. The CO_2 injection within the Midale unit has been shown to be an effective and safe method of CO_2 storage. Projects such as this are considered a key emissions reduction technology, and early demonstration and continued operation of CO_2 sequestration capability in the Midale Unit have been instrumental in Canada becoming a world leader in CO_2 sequestration technology. Since 2005 this project has injected over 5.1 million tonnes of CO_2, which would have otherwise been released to atmosphere.

1.23 tonnes

of CO_2 equivalent sequestered
for every tonne of CO_2 equivalent emitted



Emissions Reductions

In addition to CO_2 injection at Midale, Cardinal proactively implemented emission reduction projects during 2021, including conversion of pneumatic devices from high to low bleed, installation of instrument air equipment and an enclosed combustor installation to target reduction of methane and vented gas emissions. An example of one Southern Alberta site's pneumatic equipment upgrade is displayed below, alongside the associated emissions reductions from all of the 80 emissions reductions projects executed in 2021. Cardinal also secured both provincial and federal funding for emission reduction projects, including vapor recovery unit installations, to reduce vented tank top emissions. Cardinal continues to take steps to reduce methane released to the atmosphere by limiting vented emissions through proactive leak detection and maintenance. Methane has a global warming potential 25 times that of carbon dioxide, and is the primary constituent of vented natural gas, and therefore has been the target of the above described vent gas reduction projects. Cardinal is also now generating regulated carbon offsets. We are proactively engaged in the development of Carbon Markets, as these emerging markets develop due to consumer demand and a global desire to lower global greenhouse gas emissions.

PRIOR INSTALLATION:

UPGRADED SITE:

2,198 tonnes

of CO_2 equivalent emissions reductions during 2021 from executed projects

EMPLOYEE-DRIVEN SUPPORT

As part of Cardinal's charitable donation policy, a percentage of the funding earmarked for charities is directed to support civic organizations and foundations within the communities in which we operate. The program's goal is to help registered charities and organizations that directly benefit members of the local community, and to assist groups that would not normally attract funding. Some of the organizations that Cardinal supported in 2021 include:

- Medicine Hat Women's Shelter
- Brooks Food Bank
- Prairies Rose School Division
- Means for Mom's Society
- Halo Rescue
- Beaver County Victim Services
- Irma Fire Fighter's Association
- Beaverlodge Fire Department
- Slave Lake Hospital Auxiliary
- Slave Lake & District COC Riverboat Daze
- Swan River First Nation
- Sawridge First Nation
- Landon Persson Memorial Fund
- CJ Shurter School
- Tennille's Hope Kitchen
- Grizzly Firefighter's Association
- Weyburn Group Home Society
- Weyburn Humane Society
- Family Place – Weyburn
- Big Brothers/Sisters Estevan
- Inclusion Weyburn
- Envision
- Lesser Slave Watershed Council

Cardinal employee, Dave LaFrance (left) and LSWC Executive Director, Meghan Payne (right)

Employees are encouraged to submit funding requests that are ***specific to each of our operating areas.***

PERFORMANCE MEASURES

	2021	2020	2019
PRODUCTION			
Total production (boe/d)	19,090	18,442	20,319
AIR EMISSIONS			
Direct GHG emissions (10^3 tonnes CO_2e)[1]	231	160	188
GHG sequestered (10^3 tonnes CO_2e)	242	188	280
GHG emissions intensity (tonnes CO_2e per boe)[1]	0.033	0.024	0.025
Indirect GHG emissions (10^3 tonnes CO_2e)[1]	228	259	277
Flared natural gas ($10^3 m^3$/yr)	12,404	7,873	8,428
Vented natural gas ($10^3 m^3$/yr)	4,178	3,727	5,609
Methane emissions (10^3 tonnes CO_2e)[1]	66	NPR[4]	NPR[4]
SPILLS			
Reportable spills[3]	13	15	29
PERSONAL SAFETY			
Recordable injuries[4]	1	3	2
Recordable injury frequency (per 200,000 man-hours)	0.49	1.48	0.87
Lost time injuries	-	-	-
Lost time injury frequency (per 200,000 man-hours)	-	-	-
Fatalities	-	-	-
PEOPLE			
Total staff (employees & contractors)	198	190	214
Gender diversity (% female)	20%	15%	15%

1) Air emissions are calculated using industry standards and best practices in the associated jurisdiction for which our Operations take place (including Western Climate Initiative for British Columbia, Alberta Greenhouse Gas Quantification Methodologies under the TIER Regulation for Alberta, The Management and Reduction of Greenhouse Gases for Saskatchewan, and Environment and Climate Change Canada's National Inventory Report). Emissions quantification methodologies were revised during 2021 to align with several changes to these recognized industry standards, and in doing so were responsible for the vast majority of the apparent year-over-year emissions increase. As an example, the direct emissions increase from 2019 levels would have only been 1%, as compared to the above indicated 23%, had current emissions quantification methodology been applied in 2019.
2) Spills reported to a regulatory agency, as required in the jurisdiction where a spill occurs, excluding fresh water spills.
3) Recordable injuries include fatalities, permanent total disabilities, lost work cases, restricted work cases and medical treatment cases.
4) Not previously reported

ASSUMPTIONS & TERMS

Abandoned (facilities) – The permanent removal and dismantling of all surface infrastructure

Abandoned (pipelines) – A pipeline that has been physically disconnected from an active pipeline system, purged of any hydrocarbon substance or salt water, filled with fresh water, air or inert gas, and physically sealed on all ends

Abandoned (wells) – Well abandonment requires: the plugging of any completed production zones; isolation of all porous and groundwater zones; filling the well with non-saline water, or other noncorrosive fluid; assessing to ensure its long-term integrity; and finally cutting the well casing a minimum of one meter below the surface and placing a vented cap on top of the well casing

CO_2e – Carbon dioxide equivalent

Direct emissions – Are from sources that are owned and operated by Cardinal, such as fuel for engines and heaters, flares, methane venting from pneumatics and fugitive emissions

GHG – Greenhouse gas

Indirect emissions – Are GHG emissions related to sources of purchased electricity or heat

Sequester – The process of capturing and storing atmospheric carbon dioxide as a means of reducing its presence in the atmosphere and mitigating global climate change

ADVISORIES

We have taken care to ensure the information in this report is accurate. However, this report includes aspirational goals and estimates, which will differ from actual results, and is for informational purposes only. We disclaim any liability whatsoever for errors or omissions. Further, some information in this report may have been disclosed previously in other Cardinal public disclosure, and such disclosure is not intended in any way to be qualified, amended, modified or supplemented by information herein.

Material may be used within this report to describe issues for voluntary sustainability reporting that are considered to have the potential to significantly affect sustainability performance in our view and may be important in the eyes of internal or external stakeholders. However, material for the purposes of this report should not be read as equating to any use of the word in other Cardinal public reporting or filings. With this report, we hope to increase your knowledge of Cardinal and our operations. However, this report does not provide investment advice, and readers are responsible for making their own financial and investment decisions.

There is no single standard system that applies across companies for compiling and calculating the quantity of GHG, nitrogen oxide, sulphur dioxide emissions and other sustainability metrics attributable to our operations. Accordingly, such information may not be comparable with similar information reported by other companies. Our emission statistics are derived from various internal reporting systems that are generally different from those applicable to the financial information presented in our consolidated financial statements and are, in particular, subject to less sophisticated internal documentation as well as preparation and review requirements, including the general internal control environment. We may change our policies for calculating these emissions and other sustainability metrics in the future without prior notice.

In this report, Cardinal has used a number of oil and gas metrics which do not have standardized meanings and therefore may be calculated differently from the metrics presented by other oil and gas companies. Boe means barrels of oil equivalent. The term Boe may be misleading, particularly if used in isolation. The conversion ratio of six thousand cubic feet per barrel (6 Mcf: 1 Bbl) of natural gas to barrels of oil equivalent is based on an energy equivalency conversion method primarily applicable at the burner tip and does not represent a value equivalency at the wellhead. Given that the value ratio based on the current price of crude oil as compared to natural gas is significantly different from the energy equivalency of 6:1, utilizing a conversion on a 6:1 basis may be misleading as an indication of value.

This report contains forward looking statements and forward-looking information (collectively, forward-looking statements") related to future, not past events and circumstances – including those which may relate to our strategies, focus, goals, ambitions, aims, targets, plans, objectives, operations, results and financial performance. The use of any of the words "will", "may", "anticipate", "expect", "objective", "believe", "plans", "intends", "potential", "continue", "guidance", and similar expressions are intended to identify those forward-looking statements.

Forward-looking statements involve risk and uncertainty because they relate to events and depend on circumstances that will or may occur in the future including, without limitation, those risks considered under "Risk Factors" in Cardinal's Annual Information Form and which may be outside of our control. Readers are cautioned that the assumptions used in the preparation of such information, although considered reasonable at the time of preparation, may prove to be imprecise and, as such, undue reliance should not be placed on forward-looking statements. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee future results, levels of activity, performance or achievement since such expectations are inherently subject to significant business, economic, competitive, political and social uncertainties and contingencies. Many factors could cause our actual results to differ materially from those expressed or implied in any forward-looking statements made by, or on our behalf, in this report.

We have included the forward-looking statements in this report in order to provide readers with a more complete perspective on our future operations and such information may not be appropriate for other purposes. Cardinal disclaims any intention or obligation to update or revise any forward-looking statements, whether as a result of new information, future events or otherwise, except as required by law.

This report contains information from publicly available third party sources. Although management believes it to be reliable, Cardinal has not independently verified any of the data from third-party sources referred to in this document or analyzed or verified the underlying studies or surveys relied upon or referred to by such sources, or ascertained the underlying economic assumptions relied upon or referred to by such sources.

Supplemental Information Regarding Product Types

This report includes references to production. The following table is intended to provide the product type composition as defined by NI 51-101.

	LIGHT/MEDIUM CRUDE OIL	HEAVY OIL	NGL	CONVENTIONAL NATURAL GAS	TOTAL (BOE/D)
2019 average	55%	28%	4%	13%	20,319
2020 average	57%	26%	5%	12%	18,442
2021 average	54%	29%	4%	13%	19,090
Current	54%	29%	4%	13%	20,500